

Name: _____

Reg. No.: _____

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Period: 2hours



University of Aden
Faculty of Science
Department: Chemistry
Level/Semester: 2nd/1st
Examiner: Dr. Adel A.M. Saeed

Analytical Chemistry (I) Exam
2022-2023

PART I: ANSWER ALL THE QUESTIONS:

Q1) Circle clearly the correct answer (1.5 marks each):

- The pH of a $1.0 \times 10^{-8} \text{M}$ solution of LiOH is
(a) 7.02 (b) 7 (c) 8 (d) 6
- The degree of agreement between the results of repeated measurements called
(a) calibration (b) accuracy (c) precision (d) true value
- Ppm unit is
(a) $\mu\text{L/mL}$ (b) $\mu\text{g/L}$ (c) ng/L (d) $\mu\text{L/mL}$
- The most widely used of quantitative titrimetry is
(a) coulometric (b) gravimetric (c) volumetric (d) conductimetric
- Primary standards have
(a) low formula weight (b) high purity (c) hydrate water (d) high cost
- Qualitative analysis tells us what chemicals are present in
(a) a simple method (b) a sample (c) a laboratory (d) a final volume
- A liquid solution contains a liquid solute in
(a) a solid solvent (b) a liquid solvent (c) a & b (d) all are wrong
- A measured volume of reagent of known concentration reacts by the analyte, in a process called
(a) determination (b) dilution (c) titration (d) determination

Q2) Put *True* or *False* and correct the false statement (2 marks each):

0.15 g is the mass of magnesium (solute) in a body if the body weights 30 Kg and the concentration is 5.0 ppm.	
All zeros between no-zero numbers are significant.	
Analytical chemistry is the study of the composition of matter quantitatively only.	
pH (potential of hydrogen) is a numeric scale used to specify the acidity or basicity of an aqueous solution.	
Molecular mass $\times$ Molality = Equivalent mass $\times$ Normality	
E° of $\text{Ce}(\text{SO}_4)_2^{2+}$ is 1.44 Volt.	
Back-titration, residual-titration is used when the rate of reaction between the analyte and reagent is slow.	
Iodimetry is a titration method when iodine is used as the titrant.	

PART II: CHOOSE ONLY TWO (Calculator is Allowed):

Q 3) Perform the following calculations (11 marks each):

3.1- A certain experiment yielded the following data values for lead: 10 ppm, 8 ppm, 7 ppm, 11 ppm and 15 ppm.

(a) What are the mean for this experiment?

(b) Examine if 15 ppm is accepted or rejected (outlier value). $Q_{\text{tab.}} = 0.710$ at 95% CI).

3.2- Perform calculations needed to generate a titration curve for 12.00 mL of 0.200 M NaBr with 0.100 M AgNO₃ at: 0.00mL, 5.0mL, 12.0mL, 25.0mL ($K_{\text{sp}} = [\text{Ag}^+][\text{Br}^-] = 5.2 \times 10^{-13}$).

3.3- Exactly 0.8821 g of pure MgCO₃ (84.3 g/mol) is dissolved in 100.0 mL of water.

A 5.0-mL aliquot is titrated with 4.50 mL of EDTA. Calculate the molarity (M) of the EDTA and its titer (T) for MgCO₃.

3.4- Describe (by equations) how you would prepare:

(a) 2.0 L of approximate 0.5 M HClO₄ (100.46 g/mol) from the conc. reagent, which has a specific gravity of 1.67 g/mL and contains 71 % (w/w) HClO₄.

(b) 250 mL of standard 100 ppm Na⁺ solution from primary standard Na₂CO₃ (105.99 g/mol).



